Chris J. Vargo

1B15 Armory Building • University of Colorado Boulder • Boulder, CO 80303
christopher.vargo@colorado.edu • (724) 840-2973 • ORCID: 0000-0003-1733-2506
chrisjvargo.com • Google Scholar • GitHub
Current as of September 27, 2026

Education

Ph.D., Mass Communication *May 2014*
University of North Carolina at Chapel Hill

Master of Arts, Advertising & Public Relations *May 2011*
University of Alabama

Bachelor of Arts, Advertising & Public Relations *May 2008*
Pennsylvania State University

Current Academic Appointments

Associate Professor of Advertising *August 2021 – Present*
College of Media, Communication & Information
University of Colorado at Boulder

Associate Professor of Information Management and Analytics (Courtesy) *June 2021 – Present*
Leeds School of Business
University of Colorado at Boulder

Graduate Faculty Member and Instructor *January 2022 – Present*
Master of Science in Data Science
University of Colorado at Boulder

Past Academic Appointments

CMCI Director of Master of Science in Business Analytics (MSBA) *May 2018 – May 2023*
College of Media, Communication & Information
The University of Colorado at Boulder

Associate Chair of Graduate Studies *Fall 2022*
Advertising, Public Relations, and Media Design
University of Colorado at Boulder

Editor–in–Chief *December 2018 – January 2022*
The Agenda Setting Journal

Assistant Professor of Advertising *August 2016 – July 2021*
College of Media, Communication & Information
University of Colorado at Boulder

Assistant Professor of Advertising and Public Relations *August 2014 – July 2016*
College of Communication & Information Sciences
The University of Alabama

Adjunct Faculty and Graduate Research Assistant *August 2011 – May 2014*
School of Media and Journalism
The University of North Carolina at Chapel Hill

Adjunct Faculty *June – August 2011*

College of Communication & Information Sciences
The University of Alabama

Graduate Teaching Assistant *August 2009 – May 2011*
College of Communication & Information Sciences
The University of Alabama

Research

Patents

- Vargo, C., Hopp, T. (2021). Systems and Methods for Detecting Social Diversity and Inclusion. Patent Application. No. 63/133,741. CU Reference No.: CU5462B–PPA1. Filed Jan 4th, 2022.

Books

1. Vargo, C. (2024) The Computational Content Analyst: Using Machine Learning to Classify Media Messages. Routledge. https://doi.org/10.4324/9781003514237 https://www.routledge.com/The-Computational-Content-Analyst-Using-Machine-Learning-to-Classify-Media-Messages/Vargo/p/book/9781032846309
2. Shaw, D., Minooie, M., Aikat, D., & Vargo, C. (2019). Agendamelding: News, social media, audiences, and civic community. New York, NY: Peter Lang. https://doi.org/10.3726/b15023
3. Graham, G., Greenhill, A., Shaw, D., & Vargo, C. (2015). Content is king: News media management in the digital age. New York, NY: Bloomsbury.

Refereed Monographs

1. Amazeen, M., Vargo, C., & Hopp, T. (2018). Reinforcing attitudes in a gatewatching news era: Individual–level antecedents to sharing fact–checks on social media. Communication Monographs, 86(1), 112–132. https://doi.org/10.1080/03637751.2018.1521984

Refereed Articles

1. Vargo, C. J., & Barrett, B. (in press). Who gets what ads? Tail-aware metrics and the composition of mobile political exposure. *SAGE Open*. [Accepted September 23, 2026]
2. Vargo, C. J., Hopp, T., & Gangadharbatla, H. (in press). Gambling advertising and problem gambling in the United States: Multi-level evidence beyond sports-betting legal status. Policy & Internet.
3. Wang, Z., Guo, L., Wu, M., & Vargo, C. J. (in press). Imagining the world through news: A cross-national analysis of country-based issue ownership networks. *Journalism & Mass Communication Quarterly*. https://doi.org/10.1177/10776990261495869 [Accepted September 2026]
4. Vargo, C. J., Rahman, S., & Hopp, T. (2026). LLM exposure and funnel erosion in consumer journeys: Measuring journey compression and reordering in passive behavioral logs. Journal of Marketing Analytics. https://doi.org/10.1057/s41270-026-00488-w
5. Vargo, C. J., & Hopp, T. (2025). Incivility in Reddit's top political and news subreddits: Prevalence, moderation, and engagement. Social Science Computer Review. https://doi.org/10.1177/08944393251395763
6. Hopp, T., Ferrucci, P., Vargo, C. J., & Mays, B. (2025). Mainstream news media trust, countermedia attendance, and political learning. Journalism & Mass Communication Quarterly, Advance online publication. https://doi.org/10.1177/10776990251329044
7. Vargo, C. J., & Minooie, M. (2025). Breaking Agenda Setting Boundaries: A Multi-Dimensional Approach to Understanding Salience of Gun Control in the Polarized Public Sphere. Mass Communication & Society, 28(3), 530–553. https://doi.org/10.1080/15205436.2024.2333972
8. Vargo, C. J., Hopp, T., & Agarwal, P. (2024). Balancing Brand Safety and User Engagement in a Two-Sided Market: An Analysis of Content Monetization on Reddit. Journal of Current Issues & Research in Advertising, 45(2), 242–256. https://doi.org/10.1080/10641734.2023.2301621
9. Minooie, M., Taylor, B., & Vargo, C. (2023). Agendamelding and COVID-19: the dance of horizontal and vertical media in a pandemic. Frontiers in Political Science. (5)2023. https://doi.org/10.3389/fpos.2023.1021855
10. Hopp, T., Ferrucci, P., Vargo, C., & Liu, L. (2025). Is Online Textual Political Expression Associated with Political Knowledge? Communication Research, 52(1), 32–60. https://doi.org/10.1177/00936502221113808

11. Schmierbach, M., McCombs, M., Valenzuela, S., Dearing, J. W., Guo, L., Iyengar, S., Kiousis, S., Kosicki, G. M., Meraz, S., Scheufele, D. A., Stoycheff, E., Vargo, C., Weaver, D. H., & Willnat, L. W. (2022). Reflections on a Legacy: Thoughts from Scholars about Agenda-Setting Past and Future. Mass Communication and Society, 25(4), 500-527. https://doi.org/10.1080/15205436.2022.2067725

12. Vargo, C. (2022). Public "agendamelding" in the United States: Assessing the relative influence of different types of online news on partisan agendas from 2015 to 2020. Journal of Information Technology & Politics. 19(3), 284-301. http://dx.doi.org/10.1080/19331681.2021.1972893

13. Blankenship, J. & Vargo, C. (2022). The Effect of Corporate Media Ownership on Depth of Local Coverage and Issue Agendas: A Computational Case Study of Six Sinclair TV Station Websites. Electronic News, 15(3-4), 139-158. https://doi.org/10.1177/19312431211043483

14. Vargo, C. & Amazeen, M. (2021). Agenda-Cutting Versus Agenda-Building: Does Sponsored Content Influence Corporate News Coverage in U.S. Media? International Journal of Communication 15(2021), 1–22. https://ijoc.org/index.php/ijoc/article/view/17824

15. Amazeen, M. & Vargo, C. (2021). Sharing Native Advertising on Twitter: Content Analyses Examining Disclosure Practices and Their Inoculating Influence. Journalism Studies 22(7), 916–933. https://doi.org/10.1080/1461670X.2021.1906298

16. Hopp, T., Ferrucci, P., Fisher, J., Vargo, C. (2020). Exposure to Difference on Facebook, Trust, and Political Knowledge. Mass Communication & Society. 23(6), 779–809. https://doi.org/10.1080/15205436.2020.1823002

17. Guo, L., & Vargo, C. (2020). Predictors of international news flow: Exploring a networked global media system. Journal of Broadcasting & Electronic Media. 64(3), 418–437. https://doi.org/10.1080/08838151.2020.1796391

18. Hopp, T., Ferrucci, P., & Vargo, C. (2020). Why Do People Share Ideologically Extreme, False, and Misleading Content on Social Media? A Self–Report and Trace Data–Based Analysis of Countermedia Content Dissemination on Facebook and Twitter. Human Communication Research, 46(4), 357–384. https://doi.org/10.1093/hcr/hqz022

19. Vargo, C., & Hopp, T. (2020). Associations between advertisement negativity and political advertisement engagement: A computational case study of Russian–linked Facebook and Instagram content. Journalism & Mass Communication Quarterly. 97(3), 743–761. https://doi.org/10.1177/1077699020911884

20. Ferrucci, P., Hopp, T., & Vargo, C. (2019). Civic engagement, social capital, and ideological extremity: Exploring online political engagement and political expression on Facebook. New Media & Society, 22(6), 1095–1115. https://doi.org/10.1177/1461444819873110

21. Hopp, T., & Vargo, C. (2019). Social capital as an inhibitor of online political incivility: An analysis of behavioral patterns among politically active Facebook users. International Journal of Communication, 13(2019), 4313–4333. https://ijoc.org/index.php/ijoc/article/view/10523

22. Vargo, C., Gangadharbatla, H., & Hopp, T. (2019). eWOM across channels: Comparing the impact of self–enhancement, positivity bias and vengeance on Facebook and Twitter. International Journal Advertising, 38(8), 1153–1172. https://doi.org/10.1080/02650487.2019.1593720

23. Vargo, C., & Hopp, T. (2019). Attention to issues and facts: Assessing the role of need for orientation as a predictor of political news sharing on Facebook. The Agenda–Setting Journal, 3(2), 186–207. https://doi.org/10.1075/asj.18004.var

24. Guo, L., & Vargo, C. (2020). "Fake news" and emerging online media ecosystem: An integrated intermedia agenda–setting analysis during the 2016 U.S. presidential election. Communication Research, 47(2), 178–200. https://doi.org/10.1177/0093650218777177

25. Hopp, T., Vargo, C., Dixon, L., & Thain, N. (2018). Correlating self–report and trace data measures of incivility: A proof of concept. Social Science Computer Review, Special Issue "Integrating Survey Data and Digital Trace Data." 38(5), 584–599. https://doi.org/10.1177/0894439318814241

26. Vargo, C. (2018). Fifty years of agenda–setting research: New directions and challenges for the theory. The Agenda–Setting Journal, 2(2), 105–123. https://doi.org/10.1075/asj.18023.var

27. Vargo, C., Guo, L., & Amazeen, A. (2018). The agenda–setting power of fake news: A big data analysis of the online media landscape from 2014 to 2016. New Media & Society. 20(5) 2028–2049. https://doi.org/10.1177/1461444817712086

28. Guo, L., & Vargo, C. (2017). Global intermedia agenda setting: A big data analysis of international news flow. Journal of Communication, 67(4), 499–520. https://doi.org/10.1111/jcom.12311
29. Hopp, T., & Vargo, C. (2017). Does negative campaign advertising stimulate uncivil communication on social media? Measuring audience response using big data. Computers in Human Behavior, 68, 368–377. https://doi.org/10.1016/j.chb.2016.11.034
30. McGregor, S., & Vargo, C. (2017). Election–related talk and agenda setting effects on Twitter: A big data analysis of salience transfer at different levels of user participation. The Agenda–Setting Journal, 1(1), 44–63. https://doi.org/10.1075/asj.1.1.05mcg
31. Tomeny, T., Vargo, C., & El–Toukhy, S. (2017) Geographic and demographic correlates of autism–related anti–vaccine beliefs on Twitter, 2009–15. Social Science & Medicine, 191, 168–175. https://doi.org/10.1016/j.socscimed.2017.08.041
32. Vargo, C., & Guo, L. (2017). Networks, big data, and intermedia agenda–setting: An analysis of traditional, partisan, and emerging online U.S. news. Journalism & Mass Communication Quarterly, 94(4), 1031–1055. https://doi.org/10.1177/1077699016679976
33. Guo, L., Vargo, C., Pan, Z., & Ding, W. (2016). Big social data analytics in journalism and mass communication: Comparing dictionary–based text analysis and unsupervised topic modeling. Journalism & Mass Communication Quarterly, 93(2), 332–359. https://doi.org/10.1177/1077699016639231
34. Kim, Y., Gonzenbach, B., Vargo, C., & Kim, Y. (2016). First and second levels of intermedia agenda setting: Political advertising, newspapers, and Twitter during the 2012 U.S. presidential election. International Journal of Communication, 10(2016), 4550–4569.
35. Shaw, D., Mousa, I., Vargo, C., Minooie, M., & Cole, R. (2016). The agenda setting in the digital age: How we use media to monitor civic life and reframe community. Jordan Journal of Social Sciences, 9(1), 125–139.
36. Vargo, C. (2016). Toward a tweet typology: A study of brand message content types and corresponding consumer engagement on social media. Journal of Interactive Advertising, 16(2), 157–168. https://doi.org/10.1080/15252019.2016.1208125
37. Vargo, C., & Hopp, T. (2016). Social capital, political polarity, and socioeconomic status as predictors of political incivility on Twitter: A congressional district–level analysis. Social Science Computer Review, 35(1), 10–32. https://doi.org/10.1177/0894439315602858
38. Guo, L., & Vargo, C. (2015). The power of message networks: A big–data analysis of the network agenda setting model and issue ownership. Mass Communication & Society, 18(5), 557–576. https://doi.org/10.1080/15205436.2015.1045300
39. Vargo, C., Basilaia, E., & Shaw, D. (2015). Event vs. issue: Twitter reflections of major news, a case study. Studies in Media and Communication, 9, 215–239. https://doi.org/10.1108/S2050-206020150000009009
40. Shaw, D., & Vargo, C. (2014). Media and social stability: How audiences create personal community. Journalism Bimonthly [in Chinese], 6(12), 16–24.
41. Vargo, C., Guo, L., McCombs, M., & Shaw, D. (2014). Network issue agendas on Twitter during the 2012 U.S. presidential election. Journal of Communication, 64(2), 296–316. https://doi.org/10.1111/jcom.12089

Manuscripts Under Review

1. Vargo, C. J., Dobolyi, D., & Rahman, S. (2025, Oct 1). Inferring advertising demographics from behavioral data using a local LLM. Manuscript submitted for review to KDD (submitted Feb 1, 2026).
2. Vargo, C. J., & Dobolyi, D. (2025, June 4). The hidden costs of misclassified content: Advancing press sustainability and advertising integrity through improved classification [Submitted to the Journal of Interactive Marketing; revise and resubmit; revision in process]

Peer Reviewed, Edited Chapters

1. Vargo, C., & Hopp, T. (2018). Is Yik Yak a platform for political communication? Exploring college students' communication on an emergent social media platform. In N. J. Stroud & S. McGregor (Eds.), Digital discussions: How big data informs political communication (pp. 144–165). New York, NY: Routledge.

Edited Book Chapters

1. Vargo, C., & Guo, L. (2015). Exploring the network agenda setting model with big social data. In L. Guo & M. McCombs (Eds.), The power of information networks: New directions for agenda setting (pp. 55–65). New York, NY: Routledge. https://doi.org/10.4324/9781315726540

Refereed Conference Proceedings

1. Vargo, C., Hopp, T., & Gangadharbatla, H. (2025). From Ads to Addiction: The Role of Online Gambling Advertising Spend in Problem Gambling Search Trends over a Decade. Proceedings of the 58th Hawaii International Conference on System Sciences. Track: Digital and Social Media, Minitrack: Communication, Digital Conversation, and Media Technologies. https://doi.org/10.24251/HICSS.2025.288
2. Vargo, C., Masullo, G., & Hopp, T. (2024). Deciding to Delete Posts on Reddit: What Factors Influence Content Removal. Proceedings of the 57th Hawaii International Conference on System Sciences. Track: Digital and Social Media, Minitrack: Decision Making in Online Social Networks. https://doi.org/10.24251/hicss.2024.314
3. Vargo, C., Hopp, T., & Agarwal, P. (2023). Inside a Social Media Brand Safety Algorithm: A Computational Investigation of Subreddits, Toxicity, and Advertising Inventory. Proceedings of the 2023 American Academy of Advertising Annual Conference. [With graduate advisee Pritha Agarwal]
4. Vargo, C., & Hopp, T. (2023). Incivility on Popular Politics and News Subreddits: An Analysis of In-groups, Community Guidelines and Relationships with Social Media Engagement. Proceedings of the 56th Hawaii International Conference on System Sciences. Track: Digital and Social Media, Minitrack: Mediated Conversation. Maui, HI.
5. Gangadharbatla, H., Jones, V., Vargo, C., Ferrel, C., Owens, C. & Li, H. (2019). The Role of Artificial Intelligence (AI) in Advertising. Proceedings of the 2019 American Academy of Advertising Annual Conference.
6. Vargo, C., & Hopp, T. (2019). The effects of ad negativity on political digital advertising engagement: A computational case study of Russian Facebook and Instagram content. Proceedings of the 2019 American Academy of Advertising Annual Conference.
7. Hopp, T., & Vargo, C. (2016). Does negative campaign advertising stimulate uncivil communication on social media? A big data analysis. Proceedings of the 2016 American Academy of Advertising Conference, 2016, 152–153.
8. Hester, J., & Vargo, C. (2013). Social network sites and social media: A new research paradigm for strategic communication? Proceedings of the 2013 American Academy of Advertising, 2013, 177–178.

Conference Papers

1. Wang, Z., Guo, L., & Vargo, C. (2025, August). Three worlds imagined through news: A cross-national analysis of country-based issue-ownership networks. Paper accepted for presentation in the International Communication Division at the Annual Conference of the Association for Education in Journalism and Mass Communication (AEJMC), San Francisco, CA, United States.
2. Barrett, B., & Vargo, C. J. (2025, June 16). Divided by ads: Understanding partisan consumerism through online ad exposure. Paper presented in the "High-Density: Strategies in Political Advertising and Campaigns" session at the 75th Annual Conference of the International Communication Association, Denver, CO, United States.
3. Hopp, T., Ferrucci, P., Vargo, C., & Mays, B. (2024). Mainstream News Media Trust, Countermedia Attendance, and Political Learning. Paper presented at AEJMC 2024 Annual Conference, Political Communication Division, Philadelphia, PA. [With graduate advisee B. Mays]
4. Vargo, C. & Amazeen, M., (2020, August). The Competing "Content Studio" Agenda: A Large–scale Analysis of Sponsored Content in Elite U.S. Newspapers and Its Agenda Cutting Effect on Corporate News. Paper presented at AEJMC 2020 Annual Conference, Newspapers and Online News Division, Online.
5. Vargo, C., & Liu, L. (2020, August). Audience "agendamelding" in the United States: assessing the relative influence of different types of online news on partisan agendas from 2015 to 2020. Paper presented at AEJMC 2020 Annual Conference, AEJMC 2020 Theory Colloquium on Agendamelding, Online.
6. Hopp, T., Ferrucci, P., Vargo, C., & Liu, L. (2020, August). Is Facebook–Based Political Talk Associated with Political Knowledge? Paper presented at AEJMC 2020 Annual Conference, Political Communication Division, Online.

7. Vargo, C. J. (2020). Agendamelding at Scale: Assessing the Influence of Partisan, Emerging, Mainstream and Elite News Media on U.S. Audiences from 2015 to 2019. Panelist presentation at AEJMC 2020 Virtual Conference, session S066.
8. Amazeen, M., & Vargo, C. (2019, August). Sharing native advertising on Twitter: Evidence of the inoculating influence of disclosures. Paper presented at AEJMC 2019 Annual Conference, Mass Communication and Society Division, Toronto, Canada.
9. Blankenship, J., & Vargo, C. (2019, August). The effect of corporate media ownership on depth of local coverage and issue agendas: A computational case study of six Sinclair TV station websites. Paper presented at AEJMC 2019 Annual Conference, Electronic News Division, Toronto, Canada.
10. El–Toukhy, S., Vargo, C., Hopp, T., & Choi, T. (2018, February). State–level demographics and tobacco–control correlates of smoking cessation behavioral change techniques on Twitter. Paper presented at the 2018 Society for Research on Nicotine and Tobacco Annual Meeting, Baltimore, MD.
11. Hopp, T., Ferrucci, P., & Vargo, C. (2018, August). A citizen–based profile of fake news dissemination on Facebook. Paper presented at AEJMC 2018 Annual Conference, Political Communication Interest Group, Washington, DC.
12. Hopp, T., Ferrucci, P., & Vargo, C. (2018, August). Social capital, civic engagement and identity: Exploring a model for political talk on Facebook. Paper presented at AEJMC 2018 Annual Conference, Political Communication Interest Group, Washington, DC.
13. Guo, L., & Vargo, C. (2017, May). Who determines the global news agenda? A big data analysis of international news flow on the internet. Paper presented at ICA 2017, Mass Communication and Society Division (Top Faculty Paper for Journalism Studies Division), San Diego, CA.
14. Vargo, C., & Guo, L. (2016, August). A network approach to intermedia agenda–setting: A big data analysis of traditional, partisan, and emerging online U.S. news. Paper presented at AEJMC 2016 Annual Conference, Online News Division, Minneapolis, MN.
15. Vargo, C., & Hopp, T. (2016, September). Is Yik Yak a platform for political communication? Exploring college students' communication on an emergent social media platform. Digital discussions: How big data informs political communication. Paper presented at New Agendas Conference, Austin, TX.
16. Guo, L., & Vargo, C. (2015, May). The power of "issue ownership network": A big–data analysis of the 2012 U.S. presidential election. Paper presented at ICA 2015, Political Communication Division, San Juan, Puerto Rico.
17. Hopp, T., & Vargo, C. (2015, August). The effect of partisanship on changes in newspaper consumption: A longitudinal study (2008–2012). Paper presented at AEJMC 2015 Annual Conference, Mass Communication & Society Division, San Francisco, CA.
18. Kim, Y., Gonzenbach, B., Vargo, C., & Kim, Y. (2015, May). First and second levels of intermedia agenda setting: Political advertising, newspapers and Twitter postings during the 2012 U.S. presidential election. Paper presented at ICA 2015, CAT Division, San Juan, Puerto Rico.
19. Shaw, D., Vargo, C., Cole, R., & Minooie, M. (2014, October). The emerging papyrus society: How we are using media to monitor civic life, find personal community and create private identity. Paper presented at the Arab–U.S. Association for Communication Educators Annual Conference, Irbid, Jordan.
20. Vargo, C., Guo, L., Shaw, D., & McCombs, M. E. (2013, August). Network issue agendas on Twitter during the 2012 U.S. presidential election. Paper presented at AEJMC 2013 Annual Conference, Mass Communication Division, Washington, DC.
21. Vargo, C., & Shaw, D. (2013, October). When motivated reasoning mediates agenda setting: Effects in the 2012 Republican presidential primaries. Paper presented at the Annual Conference of Media and the Public Sphere, Athens, Georgia.
22. Shaw, D., & Vargo, C. (2012). The media and social stability: How audiences create personal community. Paper presented at the International Forum of Higher Education in Media and Communication, Beijing, China.
23. Vargo, C. (2012, March). Internet advertising & interactive computer services: Liability & immunity as provided by the Communications Decency Act. Paper presented at the 2012 Annual Southeast Colloquium of the AEJMC, Blacksburg, VA.

24. Vargo, C., & Arguello, J. (2012, August). The unintended consequences of "Moderate Mitt:" The ideologies of Mitt Romney & second-level agenda setting. Paper presented at AEJMC 2012 Annual Conference, Chicago, IL.
25. Vargo, C. (2011, August). Twitter as public salience: An agenda–setting analysis. Paper presented at AEJMC 2011 Annual Conference, St. Louis, MO.

Editorials (The Agenda Setting Journal)

1. Vargo, C. J. (2021). Note from the editor: Online conference follow-up. The Agenda Setting Journal, 5(1), 1–7. https://doi.org/10.1075/asj.00009.edi
2. Vargo, C. J. (2020). Note from the editor. The Agenda Setting Journal, 4(2), 171–172. https://doi.org/10.1075/asj.00007.edi
3. Vargo, C. J. (2019). Note from the editor: Key advantages of submitting work to the agenda setting journal. The Agenda Setting Journal, 3(2), 103–105. https://doi.org/10.1075/asj.00002.edi
4. Vargo, C. J. (2019). Note from the editor. The Agenda Setting Journal, 3(1), 1–2. https://doi.org/10.1075/asj.00001.edi

Invited Guest Lectures

- Vargo, C. J. (2025, April 8). Brand safety and contextual suitability with machine learning applications. Invited guest lecture for Computational Advertising (graduate level), University of Illinois Urbana–Champaign.
- Vargo, C. J. (2025, May 6). Doing new media research: Methods, strategy, and emerging directions. Invited guest lecture for "New Media Research" (graduate level), international Zoom presentation at Fudan University.
- Vargo, C. J. (2024, September 24). Agenda setting theory: Foundations and modern applications. Invited guest lecture delivered via Zoom to incoming M.S. in Journalism students, Reed School of Media and Communications, West Virginia University.
- Vargo, C. J. (2023, July 11). Computational social science across academia and industry. Invited guest lecture at the Summer Institute in Computational Social Science (SICSS–Beijing), Renmin University of China; international program featuring invited faculty from Northwestern University, Duke University, UC Davis, Purdue University, and others.
- Vargo, C. J. (2020, October 28). Big data methods for communication research. Invited guest lecture delivered via Zoom to COMM 7020 (graduate level), School of Communication and Journalism, Auburn University.
- Vargo, C. J. (2019, April 10). Invited research talk in the Information Science seminar series, Department of Information Science, University of Colorado Boulder.

Awards

- 2025, Association for Information Systems (AIS), International Conference on Information Systems (ICIS) 2025 Outstanding Reviewer.
- 2023, Hawaii International Conference on System Sciences (HICSS-56), Best Paper Award (Minitrack: Mediated Conversation, Digital and Social Media Track) for paper, "Incivility on Popular Politics and News Subreddits: An Analysis of In-groups, Community Guidelines and Relationships with Social Media Engagement."
- 2020, The University of Colorado Boulder, Provost's Faculty Achievement Award for Pre-Tenure Faculty. For the article: "The agenda–setting power of fake news: A big data analysis of the online media landscape from 2014 to 2016." Awarded $1,000.
- 2018, JMCQ Outstanding Research Article of the Year Winner for: Vargo, C., & Guo, L. (2017). Networks, big data, and intermedia agenda–setting: An analysis of traditional, partisan, and emerging online U.S. news. Journalism & Mass Communication Quarterly, 94(4), 1031–1055.
- 2017, Top Faculty Paper, Journalism Studies Division, ICA International Conference in San Diego, CA.

Grant Applications & Funds Raised

- Vargo, C. (2023) Dissecting the Anatomy of 2024 Election Disinformation: A Graph-Based Approach. Submitted to neo4j upon request. $164,143. Not funded.
- Guo, L. & Vargo, C. (2020) The agenda–setting power of fake news: The transfer of misinformation salience in a global, cross–platform information ecosystem. Submitted to Facebook. Grant duration: 2020 – 2021. Grant amount: $149,800. Submitted April 15th. Outcome: not funded.
- Vargo, C., Hopp, T., & Bradley, S. (2019). Socialcontext.ai – An ai–powered social media content strategy tool built specifically for PR agencies. Submitted to the University of Colorado's Lab Venture Challenge, August 20th, 2019. Requested $125,000. Finalist, but not funded.
- Hopp, T., Ferrucci, P. & Vargo, C. (2018) We the people: Fake news dissemination as a byproduct of citizens' civic practices. Arthur W. Page Center 'Fake News' Call for Grants, 2018. $8,500. Funded in Full.
- Hopp, T., Vargo, C., Ferrucci, P., Gondwe, G., Keegan, B., McDevitt, M., & Skewes, E. (2018). A proposal to fund the Digital Citizenship Project. de Castro Research Reward at University of Colorado, College of Media Communication & Information. $24,934. Not Funded.
- Vargo, C., & Ferrucci, P. (2018) Social Listening Studio course proposal. Payden Teaching Grant at University of Colorado, College of Media Communication & Information. $7,500. Funded.
- Vargo, C., & Hopp, T. (2018). Twitter Health Grant: Discussion productivity potential. Twitter $50,000. Submitted on April 12th. Selected as top 5 finalist amongst 239 submissions. Not Funded.
- Vargo, C. (2016). National Institutes of Health (NIH) (award number/mechanism: *[add grant/award ID here]*). Health disparity, tobacco use and socioeconomic factors: A longitudinal study of Twitter users across the United States. $25,000. Funded.
- MacCall, S. L., McMillan, D. J., Vargo, C. J., Bradley, S. B., & Aversa, E. A. (2016). Knight Foundation's news challenge for libraries: How might libraries serve 21st century information needs?: Efficiency, integration, interoperability: A 21st century approach to organizing sports digital assets for all libraries – pre–budget grant submission. Not funded.
- Hopp, T., Vargo, C., & Ames, B. (2015). Development of a scalable, multimodal identifier of malware propagation likelihood for NSF SaTC New Collaboration EAGER. Returned on March 27, 2015. Not selected to submit full proposal.
- MacCall, S. L., Vargo, C. J., Bradley, S. B., & Aversa, E. A. (2015). National Science Foundation – Small Business Innovation Research (SBIR) Phase I Grant: Development of a novel digital asset organizing method in sports – $225,000 ($74,925 sub–award to University of Alabama). Not funded.
- Vargo, C. (2015). The University of Alabama – Research Grants Committee Area B Award. Understanding how issues and attributes are created and evolve on social media using big data and Exponential Random Graph Models (ERGMs). $6,000. Funded June 1, 2015.

Teaching

For Credit Coursera Specialization – Text Analytics in Marketing

This three-credit sequence of courses was created for the University of Colorado's Master of Science in Data Science, a fully online graduate degree program offered via Coursera. Each course applies leading computer science methods to solve marketing problems using Python. Students receive conceptual overviews of methods, and dive into real-world datasets through instructor-led tutorials. Each course culminates with a major project. Last refreshed with new content Fall 2025.

Supervised Text Classification for Marketing Analytics *February 2022 – Present*

This foundational course establishes comprehensive mastery of traditional text classification methodologies applied to authentic marketing datasets. Students develop advanced Python programming skills specifically tailored for text analytics, implement sophisticated keyword analysis and content analysis techniques with intercoder reliability validation, and master multiple machine learning algorithms including Support Vector Machines, Naive Bayes, Random Forest, and ensemble methods.

Advanced LLM Classification Systems and Production Deployment *April 2022 – Present*

This cutting-edge course provides comprehensive mastery of state-of-the-art Large Language Model classification systems for enterprise marketing applications. Students gain deep expertise in transformer architectures (GPT, BERT, LLaMA variants), advanced parameter-efficient fine-tuning techniques (LoRA, QLoRA, Unsloth), high-performance inference optimization (vLLM, quantization), and production-scale deployment strategies. The curriculum emphasizes hands-on implementation using industry-leading tools including Hugging Face Transformers, vLLM for inference acceleration, comprehensive evaluation frameworks, bias detection methodologies, and ethical AI governance for responsible deployment.

Topic Modeling and Network Analysis for Marketing Insights *April 2022 – Present*

This advanced course integrates cutting-edge unsupervised learning methodologies combining sophisticated topic modeling techniques with comprehensive network analysis for strategic marketing intelligence. Students master state-of-the-art topic discovery methods including BERTopic with transformer embeddings, Non-negative Matrix Factorization (NMF), Top2Vec, and advanced clustering algorithms applied to authentic Amazon product reviews and Twitter social media data.

Coursera Specialization – Digital Advertising Strategy

This three-course (non-credit) sequence is fully online, asynchronous and offered via Coursera. Students receive conceptual overviews of advertising platforms, and dive into assignments that exercise critical thinking about digital advertising strategy. Last refreshed Summer, 2024.

Search Advertising *December 2019 – Present*

This course shows small businesses how to create and execute search campaigns on Google Ads Search (formally AdWords). Through an introductory overview, students are guided through the official Google Ads Search training materials where they will ultimately earn an official Google Ads Search Certification. Beyond Google materials, practical campaign creation and optimization best practices and exercises are provided by real experts.

Social Media Advertising *December 2019 – Present*

This course unpacks small business use cases of Facebook, Instagram and Twitter advertising. From basic campaigns, to advanced techniques including lookalike modeling and audience retargeting, this course shows how to effectively advertise on three major social media platforms.

Native Advertising *2019 – Present*

This course outlines a case study where a small travel startup used native advertising to drive hotel sales. Execution strategies for a successful, no–creative native campaigns are laid out, including: gathering existing news coverage, ethical content seeding, and content generation.

Traditional Courses

Quantitative Methods (PhD level) *Fall 2024, Fall 2025*

Introduces graduate students to quantitative social scientific research in the communication sciences. Topics covered will include – but are not limited to – the following: variables; research questions and hypotheses; causality; measurement; research design; descriptive statistics; and inferential statistics.

Social Media Listening *Fall 2021, Fall 2024, Fall 2025*

This course covers the various applications of social listening as well as the ethics of using social media data. Students use cutting edge social media tools to tackle real-world social listening tasks. This class focuses on social media research activities, including analytics, listening and engagement. Social media metrics are also introduced and discussed.

Digital Advertising *Fall 2018, Fall 2019, Fall 2020, Spring 2022, Spring 2023*

Part of the Leeds MS in Business Analytics, Marketing Track. Covers traditional and emerging digital advertising

platforms. Students execute ads at an advanced level and use leading analytic tools to assess advertising performance. Core advertising platforms covered include search, display, social media, native advertising, mobile and programmatic.

Advanced Advertising Analytics *Spring 2019, Spring 2020, Spring 2021*

This course tackles advanced advertising and marketing analytics through three advanced methods aimed at solving these problems: time series analysis, topic modeling via unsupervised machine learning, and network analysis. Includes a deep dive into the leading computer science methods aimed at solving these methods using Python and other open–source packages. Students walkthrough conceptual overviews of the methods, and dive in to real–world datasets through instructor–led tutorials.

Principles of Advertising *Fall 2016, Spring 2017, Maymester 2017, Summer A 2017, Fall 2017*

Introduction to the foundations, nature, and practice of advertising. Commercial aspects of communications. Ethical, legal and social responsibility aspects of advertising.

Content Analytics *Fall 2016, Spring 2017, Fall 2017, Fall 2018, Spring 2019, Fall 2019, Fall 2021, Spring 2022, Fall 2022, Spring 2023*

This class focuses on measuring and evaluating the effectiveness of strategic communication content (i.e., ads, webpages, and social media). It also provides an overview of metrics and best practices (e.g. KPIs) by medium.

Social Media Strategy *Summer 2016*

Provides students with the practical, theoretical and analytical knowledge and skills required to successfully develop, monitor and execute digitally based and social media campaigns. Students will acquire a skill set based on the demands of current industry practice.

Independent Study *Spring 2015, Fall 2015*

Brands & Virality. Independent study with students interested in brands' use of social media and how that use leads to diffusion of messages.

Social Media Persuasion *Fall 2015*

Master's level course on the practice of creating, writing, editing and producing persuasive communication for advertising and public relations.

Public Relations Writing *Fall 2014, Spring 2015, Fall 2015, Spring 2016*

Theory and practice involved in creating messages, including planning, writing, editing, production and evaluation.

Design Application *Fall 2010, Spring 2011, Summer 2014, Summer 2015*

An introductory skills course for advertising and public relations majors and minors using Adobe Photoshop, Illustrator and InDesign. Students develop skills to design logos, ads, brochures, newsletters and other basic APR documents.

Visual Communication *Summer 2011, Spring 2015*

Theory, concepts and aesthetics applied to commercial persuasion in advertising and public relations. Particular attention is given to the application of course content in desktop publishing.

Advertising Media *Spring 2014*

200–level course in media planning, buying and selling. Emphasized focus in buying and planning for social media.

Agenda–Setting Theory Seminar *Spring 2013*

400–level course on the current and past works in agenda setting. The goal of the course is to have each student conduct a research study in agenda setting and complete it by semester's end. Co–instructed with Dr. Donald

Shaw.

Introduction to Multimedia Design *Summer 2012, Summer 2013*

Entry–level course in multimedia storytelling that includes modules on theory, the profession, design, content gathering, editing, programming, publishing and usability.

Creative Copy & Communication *Fall 2012*

Junior and senior–level course that reviews creative theory from social science research and applies thinking to advertising campaigns. Students brainstorm and execute multi–platform campaigns, while reviewing creative best practices.

Service

Service to the University of Colorado Boulder

- Faculty Representative, OIT Cabinet AI Steering Committee, Spring 2026 — Present
- Faculty Representative, OIT Cabinet OpenAI Steering Committee, Spring 2026 — Present
- Member, Boulder Faculty Affairs Campus Operations & Resources Committee (CORC), Fall 2025 — Present
- Member, Strategic Facilities Visioning Committee, Fall 2018 — 2024
- Faculty Partner, Office of Academic and Learning Innovation, Fall 2018 — Present

Service to the College of Media, Communication & Information

- Faculty Council Member, Fall 2021 — Present
- Personnel Committee Member, Fall 2021 — Spring 2023
- Director of Master of Science in Business Analytics (MSBA), Spring 2018 — 2023
- Co–organizer for Business Analytics Meetup Lecture Series, Fall 2018 — 2022

Service to the Advertising, Public Relations and Media Design Department

- Research roundtable presentation (30 minutes), CMCI APRD Journalism, University of Colorado Boulder, November 5, 2025. Demonstrated workflows for pulling, downloading, and extracting large-scale digital trace data using Meltwater, with examples for graduate students and junior researchers.
- APRD/Journalism Research Colloquium Coordinator, AY 2025
- Graduate Affairs Committee, AY 2023, AY 2024
- Associate Chair of Graduate Studies, Fall 2022
- Search Chair, 3 tenure-track lines in Strategic Advertising, Fall 2022
- Member, Primary Unit Evaluation Committee, Fall 2021
- PUEC mid-tenure review (Mia Wang), Spring 2026
- Search Chair, Scholar in Residence of Digital Advertising Strategy, Fall 2021 — Spring 2022
- APRD Chair Search Committee Member, Fall 2019 — Spring 2020
- APRD Data Analytics Instructor Search Committee Chair, Spring 2019
- Chair and Member of Executive Committee, Spring 2017 — Fall 2017
- Industry Partnerships Committee Member, Summer 2017 — Spring 2018

- AD + PR in Paris Summer 2018 Faculty–led Global Seminar Co–Director, Fall 2017
- Advised Internships (Summer 2017: 7, Fall 2017: 2, Summer 2018: 3, Spring 2019: 2, Summer 2019: 1, Fall 2019: 1, Fall 2020: 1, Spring 2021: 1; Spring 2022: 1)
- Directed Independent Studies (Fall, 2017: 1 Graduate, 1 Undergraduate; Spring, 2022: 1 Graduate)
- PR Search Committee Member, 2016

Service to the Academy

- Discussant, Communication Theory & Methodology Scholar-to-Scholar (Poster) Refereed Paper Session, AEJMC 2020 Virtual Conference.
- Guest Co–Editor, Special Issue: Agenda Manipulation in a Platformized Attention Economy: Building, Cutting, and the Governance of Salience, Frontiers in Political Science, Accepted Oct 2025. (with Lei Guo, Milad Minooie). [Call for papers announced, accepting submissions]
- Guest Co–Editor, Special Issue: Revisiting Agenda-Setting Theory in the Age of AI, Communication and Change, Acknowledged in print. (with Lei Guo).
- Editor, the Agenda Setting Journal, December 2018 — 2024.
- External Reviewer for Tenure & Promotion, University of Southern California (USC) Annenberg School for Communication and Journalism, 2022 (1 case).
- Guest Editor, the Agenda Setting Journal, Spring 2018 — Summer 2018.

Dissertation Committees

- Primary advisor (through November 7, 2024), Pritha Agarwal, University of Colorado Boulder, 2022 – 2024
- Primary advisor (Spring 2026 — Present), Shahed Rahman, University of Colorado Boulder
- Member, Yiyan Zhang, Boston University, Spring 2020 – Summer 2021

Thesis Committees

- Matt Brockman, University of Arizona, Spring 2020
- Tom Arenberg, University of Alabama, Fall 2015 – Fall 2017

Manuscript Reviewer

- American Academy of Advertising (AAA), 2017, 2019, 2022
- AEJMC National Conference, Ad Division, 2016, 2017, 2019, 2020
- AEJMC National Conference, Newspaper and Online News Division, 2020, 2021
- AEJMC National Conference, PR Division, 2015
- Big Data & Society, 2018
- Chinese Journal of Communication, 2020
- Communication Methods and Measures, 2019, 2020
- Communication Research, 2020, 2021
- Computers in Human Behavior, 2018
- Emerald Studies in Media and Communication, 2016
- Harvard Kennedy School (HKS) Misinformation Review, 2021
- International Journal of Communication, 2016 – 2022
- International Journal of Press and Politics, 2015
- International Journal of Public Opinion Research, 2021
- Information, Communication & Society, 2020
- Journalism & Mass Communication Quarterly, 2015 – 2021
- Journalism, 2015, 2021
- Journalism Studies, 2019
- Journal of Broadcasting & Electronic Media, 2019, 2022

- Journal of Communication, 2015 – 2017, 2019 – 2022
- Journal of Information Technology & Politics, 2015 – 2016, 2018 – 2023
- Journal of Interactive Advertising, 2018 – 2019
- Journal of Marketing Communications, 2017
- Journal of Public Relations Research, 2019
- Mass Communication & Society, 2017, 2019, 2020
- New Media & Society, 2015, 2017, 2019–2022
- Political Communication, 2014, 2016, 2020 – 2021
- Social Media + Society, 2018
- Social Science Computer Review, 2015, 2019 – 2021

Professional Consulting

- Technical expert, Spiro Harrison & Nelson, *Lamothe v. Decentral Life, Inc. et al.*, No. 1:23-cv-03251 (D. Colo.), 2025–2026. Expert report filed as Doc. 86-1; MAU/KPI reconstruction, SQL evidence, platform analytics, and external traffic triangulation.
- Technical expert, Holland & Knight LLP/APCIA, Colorado HB23-1004 language-access insurance advertising, 2024. Advertising, translation-feasibility, and policy-impact analysis.
- Technical expert, The McCallum Law Firm, P.C. / Bakos Law, *Fernandez v. Tequilas Inc. et al.*, No. 1:23-cv-01382 (D. Colo.), 2025–2026. SEMrush, traffic, earned-media, advertising-value, and brand-use evidence.
- Marketing expert, SBIGlobal and Cooper Ed Law, 2025–present. Google Ads, search analytics, field-seller analytics, survey scoring, transcript review, and coaching outputs.

Prior Professional Experience

- Founder and CEO, socialcontext LLC (D.B.A. socialcontext.ai), June 2019 – 2025. Built Impact Index with WPP/GroupM Mindshare for Tyson; Ad Age covered four-class editorial-impact scoring, an under-10 tech team, and weekly audits targeting 90% precision.
- Digital and Social Media Creative, Porter Novelli, Atlanta, GA, May – August 2010
- Digital and Media Coordinator for the Digital Marketing Department, Epic Records/Sony Music, New York, NY, August – December 2008
- Postgraduate Internship, Digital Marketing Department, Columbia Records/Sony BMG, New York, NY, June – August 2008
- Public Relations Coordinator, FOX/Mark Burnett/DreamWorks, Los Angeles, CA, May – August 2007
- Public Relations Intern, College of Engineering, The Pennsylvania State University, State College, PA, August 2007 – May 2008